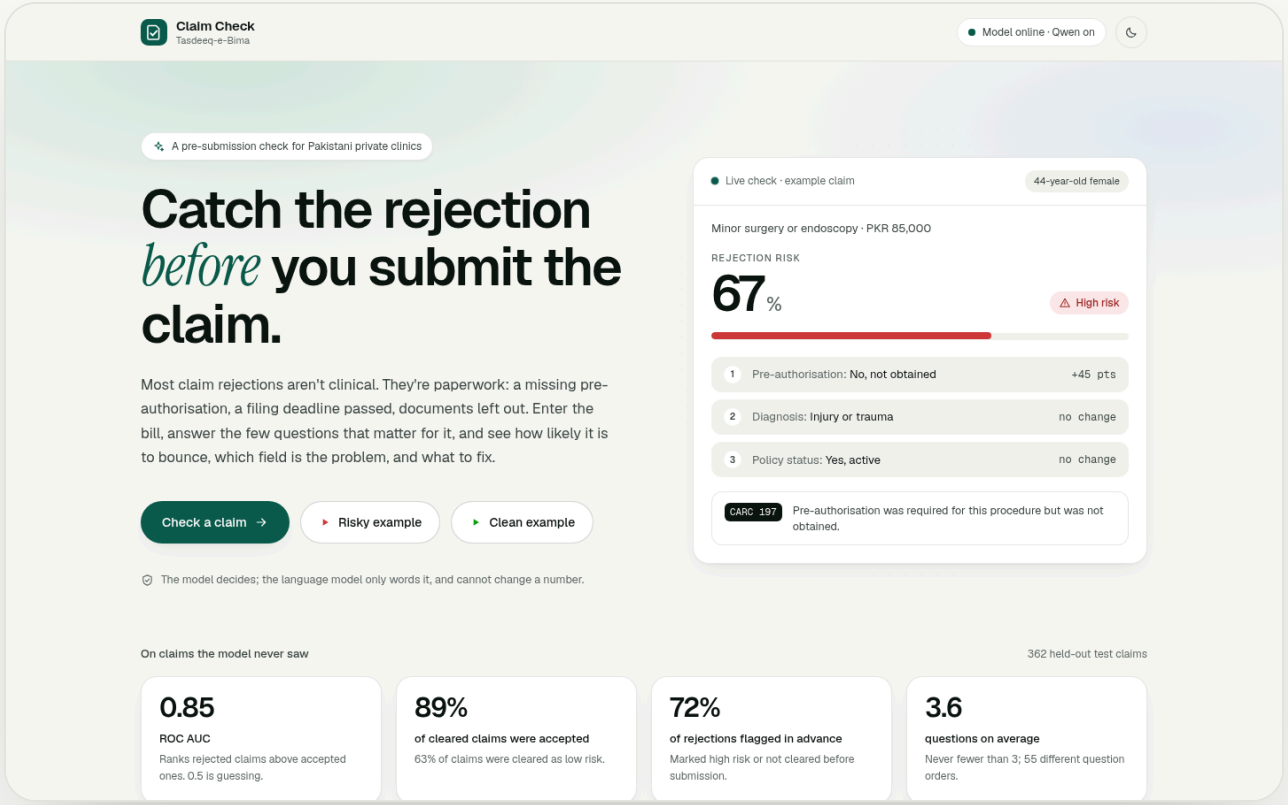
BANO QABIL AI HACKATHON

Claim Check

Catch the insurance-claim rejection *before* the claim is sent.



Mirza Ahsan Baig · Safee Akmal

Tasdeeq-e-Bima

THE PROBLEM

Claims get rejected for paperwork, not medicine

- Weeks after treatment, the claim bounces: **no pre-authorisation**, a **missed deadline**, **missing documents**
- The clinic reworks it and waits again for its money
- Small clinics can’t absorb that — they stop taking insured patients

PRE-AUTHORISATION

Insurer approval needed **before** a planned treatment. Miss it and correct care can still go unpaid.[6]

Sources: [6] American Medical Association

THE SCALE

A global problem, hardest where insurance is new

UNITED STATES

15%

of claims to private insurers denied first time[1]

UNITED STATES

$25.7B

spent by providers fighting denials in 2023[1]

UNITED STATES

86%

of denials potentially avoidable[2]

INDIA

71%

of health-claim value actually paid, FY 2023-24[3]

SEHAT CARD SCHEME

5–8 months

payment waits, blamed on incomplete documents[4]

KENYA

600+

rural private hospitals stopped accepting the national cover[5]

Sources: [1] Premier Inc. · [2] Becker's Hospital Review · [3] Outlook Business · [4] Aaj News · [5] Business Daily Africa

OUR SOLUTION

A spell-checker for insurance claims

1 **Photograph the papers**

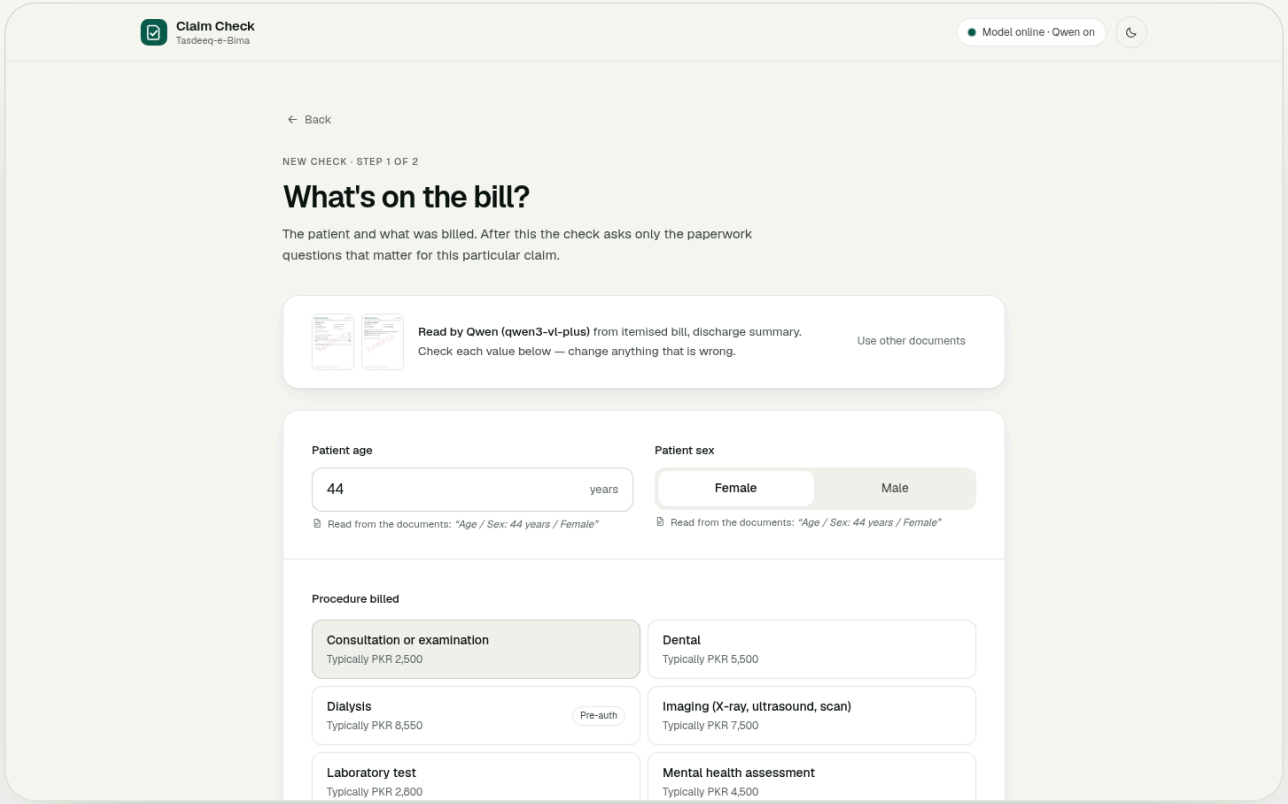
Qwen reads them; the clerk confirms

2 **Answer what they can’t**

Three or four questions, chosen for this claim

3 **Fix before sending**

Chance of rejection, the cause, the fix



BUSINESS CASE

Everyone in the claim wins

CLINICS

- Paid sooner
- Less rework
- Predictable cash flow

INSURERS & TPAS

- Cleaner claims
- Fewer appeals
- Lower admin cost

PATIENTS

- Clinics keep accepting their insurance
- Care isn’t held up by paperwork

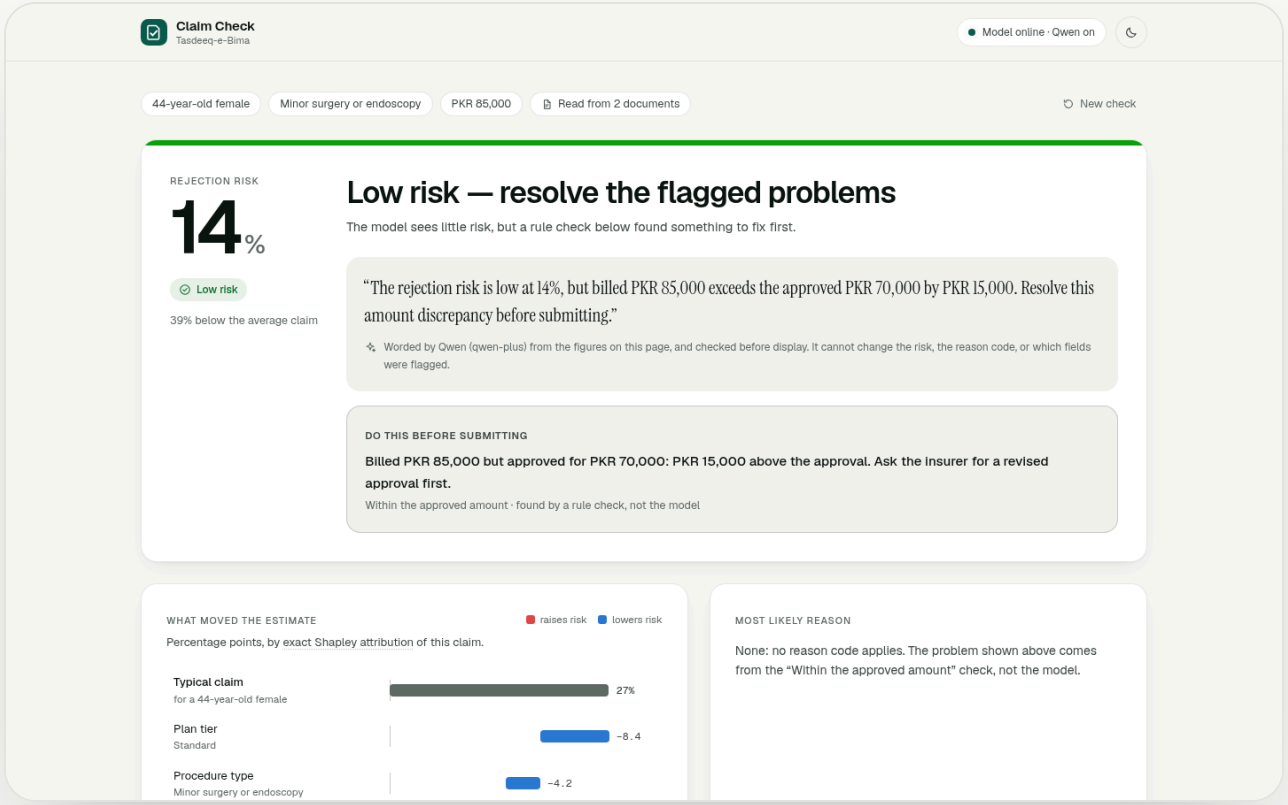
How it could earn (proposed): per-clinic subscription · licence to TPAs for their provider panel · integration with national e-claims schemes

HOW IT WORKS

Reads the papers, asks the rest, says what to fix

- Fills in what is printed, then asks what the papers can't settle
- Cross-checks the papers: a missing approval, an approval below the bill
- Risk updates after every answer; stops once settled
- The cause, the official rejection code, the fix — and a one-page printout

3.6 questions on average · 55 different question orders



RESPONSIBLE AI

Qwen reads and explains. The model decides.

QWEN-VL · READS

Fills in only what is printed; the clerk confirms every value

MODEL + RULE TABLES · DECIDE

The chance of rejection, the verdict, the reason, the checks

QWEN · EXPLAINS

One plain sentence from the finished result

- Pre-authorisation is **never** taken from a document — it is always asked
- Any reply quoting a number it wasn't given is **discarded** for a fixed template
- No network? Type the bill; the template still works — the demo never breaks

RESULTS

Tested on claims it never saw

89%

of claims it cleared were accepted

72%

of rejections flagged before submission

56%

of “high risk” claims were rejected (average 24%)

0.85

AUC with every answer known (0.76 mid-check)

362 held-out simulated claims, used once. Verdict thresholds were fitted on a separate split, not chosen by hand.

HONEST ABOUT IT

Limits today, and what comes next

WHAT IT DOES NOT DO

- Train on real claims — none are public; records are simulated, rules and codes are real
- Replace staff — it is a second check
- Read every photo perfectly — each value shows its quote to check
- Submit claims or learn by itself
- Protect real patient data yet — no login

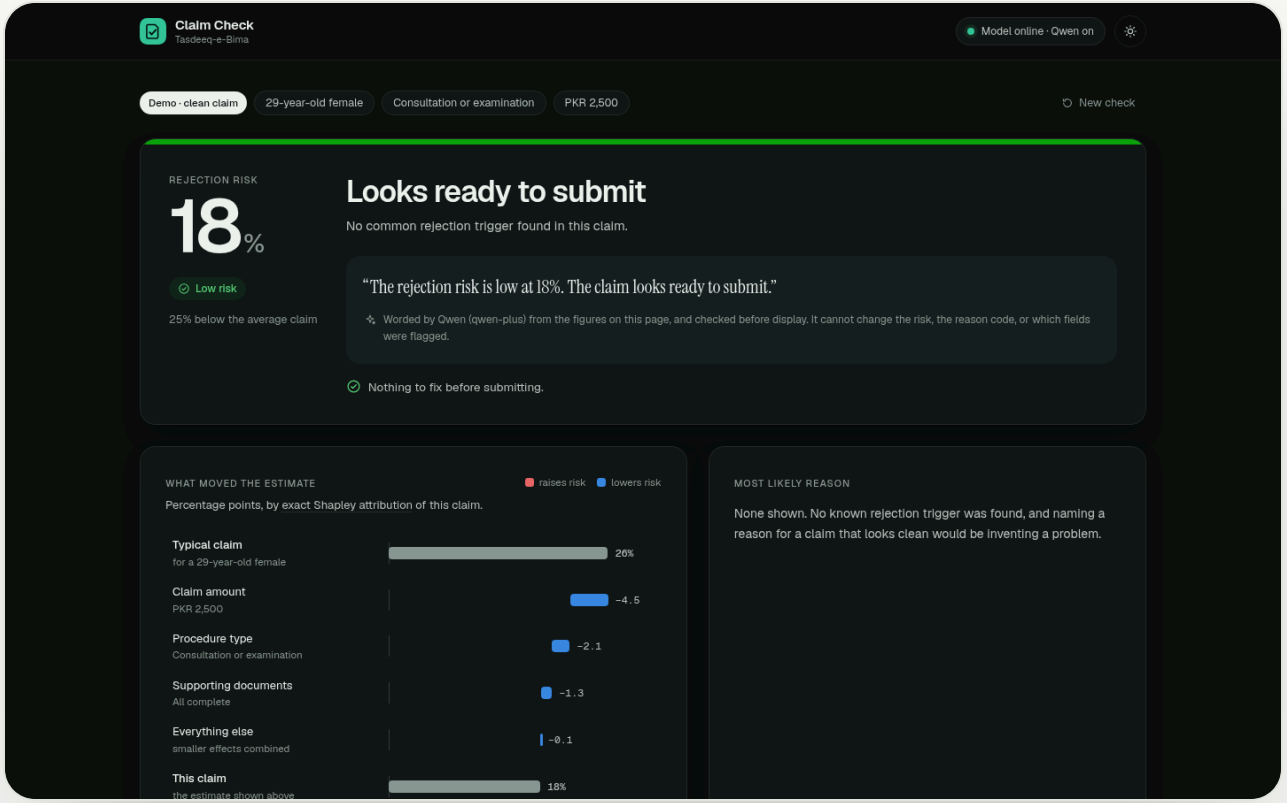
WHAT WE'D BUILD NEXT

- Pilot with a clinic or TPA on real claims
- Per-insurer rules and models
- Urdu and regional-language interface
- Login, encryption, links to billing systems

CLAIM CHECK · TASDEEQ-E-BIMA

Catch the rejection before you submit.

We're looking for a **partner clinic or TPA** to pilot with real claims.



Mirza Ahsan Baig · Safee Akmal

Thank you

SOURCES

Where the figures come from

1. Premier Inc., “Claims Adjudication Costs Providers $25.7 Billion”, Feb 2025 (2023 data)
   premierinc.com/newsroom/blog/claims-adjudication-costs-providers-25-7-billion
2. Becker’s Hospital Review, on the Change Healthcare 2020 Revenue Cycle Denials Index (102M claims, 1,500 hospitals)
   beckershospitalreview.com/finance/86-of-denials-are-potentially-avoidable-strategies-to-better-prevent-manage-denials
3. Outlook Business, on the IRDAI Annual Report 2023-24, 30 Dec 2024
   outlookbusiness.com/economy-and-policy/health-insurers-disallowed-claims-worth-rs-15100-crore-during-fy24
4. Aaj News, “More pain than relief for KP health card holders” (Baker Tilly audit), 22 Nov 2022
   english.aaj.tv/news/30304857
5. Business Daily Africa, “Rural private hospitals reject SHA cover over Sh22bn debt”, 20 Feb 2025
   businessdailyafrica.com/bd/corporate/health/rural-private-hospitals-reject-sha-cover-over-sh22bn-debt-4934984
6. American Medical Association, 2024 Prior Authorization Physician Survey
   ama-assn.org/press-center/ama-press-releases/ama-survey-indicates-prior-authorization-wreaks-havoc-patient-care

Our own results: models/metadata.json and docs/model-card.md in the project repository (3,612 simulated claims, test split of 362).